

LOCALIZATION IN THE COMMUTANT, STAGE 11:

tilings, Bratteli diagrams and singular foliations.

Depth as a parabolic renormalization, the two tiers as one flow,
and the walls of a RELU network as a stratified foliation

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Abstract

We read eight papers: the tiling-groupoid and Bratteli-diagram papers of Bellissard–Julien–Savinien and Julien–Savinien, the bi-infinite Bratteli diagrams and translation surfaces of Putnam–Treviño, the measure-free Koopman–von Neumann spectral triples of Giannakis–Montgomery, Paul’s semiclassics beyond Ehrenfest time, and three papers on singular foliations (Francis; Fischer and Laurent-Gengoux; Louis). We use them to place the bias-free He-initialised RELU network with Gaussian input, and the two-tier (localization over network) framework of stages 9 and 10, inside tiling and foliation theory. The results are these. (i) At a fixed input the network is a weighted Bratteli diagram on the layered neuron set. Forward activations and backward sensitivities are its two harmonic states, and Putnam–Treviño’s conservation of $\sum_v \nu_r \nu_s$ is backpropagation: the output equals $\langle h_l, \partial_{h_l} f \rangle$ at every layer. The RELU rescaling symmetry is the diagonal (Teichmüller-type) flow, and Rauzy–Veech induction is itself a gated linear network whose activation tiling is the partition into Rauzy cylinders. (ii) Depth is a *parabolic* renormalization. The folding map of the arc-cosine kernel is the germ $\theta - \theta^2/3\pi - \theta^3/18\pi^2 + \dots$, with Fatou coordinate $3\pi/\theta + \frac{3}{2} \log \theta$ and iterative residue $3/2$. Hence $\theta_l = 3\pi/(l + \frac{3}{2} \log l + 4.39 + o(1))$ from orthogonal inputs. At $l = 16$ this gives 0.384 against the exact 0.378, whereas the stage-10 law $3\pi/l$ gives 0.589. (iii) The two tiers are one flow. The unresolved gate variance at depth l and localization time t depends only on the Fatou time $\tau = \Phi(\theta_0(t)) + l - 1$. Localization deforms the state, a layer shifts the diagram, and the Fatou coordinate is the renormalization time, exactly as in Putnam–Treviño’s description of the Teichmüller flow. One layer is worth $\sqrt{2}/3\pi$ of $\sqrt{1+t}$, and measured width-512 data collapse on τ except at the last layer. (iv) The walls form a stratified singular foliation whose leaves are the faces of the tiling. Crossings of walls of one layer are normal crossings with abelian isotropy. A deep wall crossing a wall that feeds it is bent; its formal transverse model is a three-line arrangement whose isotropy algebra is the non-abelian $\mathfrak{aff}(1)$, and the bending is the gated transport coefficient at which Price’s theorem gives birth to the third cumulant. Faces are contractible and every wall has a global defining function, so outer holonomy is trivial and the noncommutativity is all in this isotropy. (v) The upper tier is a semiclassical deformation: averaging over Gaussian balls is a coherent-state (Husimi) symbol with an exact $\sqrt{\hbar}$ boundary layer at every wall. At the target scale there are $n/4$ unresolved gates per layer, so the infinite-width description is a kinetic (van Hove) limit rather than a semiclassical one. (vi) The mean is a border functional on the codimension-one faces, written as a supertrace pairing of the wall flip with the normal derivative. Every claim is labelled proved, known, conjectural or an analogy. The identities are checked by `scripts/verify_stage11.py`.

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1 The placement in one page

Notation as in stage 10: $W_l \in \mathbb{R}^{n \times n}$ with i.i.d. $N(0, 2/n)$ entries, $h_0 = x \sim \gamma = N(0, I_n)$, $z_l = W_l h_{l-1}$, $h_l = \text{relu}(z_l)$, gates $g_l = \mathbf{1}\{z_l > 0\}$, $D_l = \text{diag}(g_l)$; for an output functional $f = c^\top h_L$ put $\delta_l = \partial f / \partial h_l$. The activation tiling at depth l is the partition of input space by $(g_1, \dots, g_l)$; its walls are $Z_{l,a} = \{z_{l,a} = 0\}$. The upper tier is the family of laws $N(m, \hbar I)$ (or Eldan's path with $\hbar = 1/(1+t)$), and $F(m, \hbar) = \mathbb{E}f(m + \sqrt{\hbar}Z)$.

1. The cocycle (Section 3). At a fixed input the network is a weighted Bratteli diagram on the layered neuron set, with edge matrices $E_l(x) = D_l(x)W_l$. Forward activations are a left-harmonic state, backward sensitivities a right-harmonic state, and Putnam–Treviño's invariant $\sum_{v \in V_l} \nu_r(v) \nu_s(v)$ is $\langle h_l, \delta_l \rangle = f(x)$ at every layer (Theorem 3.2). Neuron i of layer l is a rectangle of width $h_{l,i}$ and height $\delta_{l,i}$ (its gradient $\times$ input attribution), each layer is a zippered-rectangles presentation of the same area, and the ReLU rescaling symmetry is the diagonal flow that fixes every rectangle's area. Moving the input across a wall is a rank-one elementary move of the cocycle, the analogue of a Rauzy move. Conversely, Rauzy–Veech induction is a gated linear network: one gate per step, its activation regions are the Rauzy cylinders, and its Jacobians are the Rauzy–Veech cocycle (Proposition 3.3).

2. Depth is parabolic (Section 4). The pair angle obeys $\theta_l = F(\theta_{l-1})$ with $F(\theta) = \theta - \theta^2/3\pi - \theta^3/18\pi^2 + O(\theta^4)$, a parabolic germ. Its Fatou coordinate is $\Phi(\theta) = 3\pi/\theta + \frac{3}{2} \log \theta + O(\theta)$ (iterative residue $3/2$), and $\Phi \circ F = \Phi + 1$. This replaces the stage-10 folding law $3\pi/l$, which is 56% off at $l = 16$, by $\theta_l = 3\pi/(l + \frac{3}{2} \log l + 4.39 + o(1))$, which is 1.6% off (Theorem 4.1). The He-critical network is a Farey-type (additive, parabolic) renormalization, not a Gauss-type (hyperbolic) one. The infinite sum $\sum_l \theta_l$ diverges, so at infinite width two distinct inputs disagree on gates at infinitely many layers.

3. The two tiers are one flow (Theorem 4.3). The unresolved gate variance at depth l and localization time t is a function of the Fatou time $\tau = \Phi(\theta_0(t)) + l - 1$ alone. In Putnam–Treviño's words, the Teichmüller flow is a deformation of the state followed by a shift of the diagram: here the localization deforms the state, a layer shifts the diagram, and the Fatou coordinate measures

the renormalization time. One layer of depth is worth $\Delta\sqrt{1+t} = \sqrt{2}/3\pi \approx 0.150$. The measured width-512 disagreement fractions of stage 10, sorted by τ , fall on one curve except at the last layer.

4. The walls are a stratified singular foliation (Section 5). Its leaves are the open faces of the tiling. Walls of one layer cross normally, so the isotropy there is abelian (b -calculus, Debord, Helffer–Nourrigat cone equal to the log cotangent). A deep wall crossing a wall that feeds it is bent by the gated transport coefficient $T = \partial z_{l,a}/\partial h_{k,b}$. Its formal transverse model is the three-line arrangement $y_1 y_2 (y_2 + T y_1) = 0$, whose isotropy algebra is $\mathbf{aff}(1)$ with $[E, \vartheta] = \vartheta$. The linear isotropy drops from dimension 2 to 1 (Theorem 5.2). The bent crossings are where Price’s theorem gives birth to the third cumulant: with orthogonal first-layer rows, $\kappa_3(z_{2,a}) = c_3 \sum_b T_{ab}^3 |w_b|^3$, $c_3 = (\pi+2)/(2\pi)^{3/2}$. Faces are contractible and walls have global defining functions, so by Fischer–Laurent–Gengoux and Francis there is no holonomy. The wall foliation’s noncommutativity is purely isotropic and sits exactly on the memory.

5. The upper tier is a semiclassical deformation (Section 6). $N(m, \hbar I)$ is a coherent state and $F(m, \hbar)$ the network’s Husimi symbol. Near a wall, $F(m, \hbar) - f(m) = \sqrt{\hbar} \kappa \psi(d/\sqrt{\hbar})$ up to exponentially small terms, with $\psi(u) = \varphi(u) - |u| \Phi_{\mathcal{N}}(-|u|)$. So F is smooth on the parabolic blowup of the walls, and corners need iterated blowups along the face poset (Theorem 6.1). The target $\hbar = 1$ is deep in the quantum regime ($n/4$ unresolved gates per layer). The infinite-width (moment-chain) description is a weak-coupling, high-density kinetic limit. In Giannakis–Montgomery’s language, chains are Fock truncations, deep layers approach the commutative (classical) quotient, and the target is a kernel mean embedding whose estimation error is a maximum mean discrepancy.

6. Borders (Section 7). The depth tower carries two Bratteli diagrams. The tile diagram refines and the fiber diagram folds; the fibers of h_l play the role of Bellissard–Julien–Savinien’s supertiles. Border dimension is the codimension of the face containing a point. The mean is a border functional, $\mathbb{E}f = (2\pi)^{-1/2} \sum_{\text{codim-1 faces}} \kappa_F \alpha_F$, equivalently the supertrace pairing of the wall flip with the normal derivative (checked to eight digits in two dimensions).

2 What the eight papers do

Bellissard–Julien–Savinien [3]. A tiling of finite local complexity has a hull Ω (the closure of its translates) and a canonical transversal Ξ (tilings with a puncture at the origin), a Cantor set carrying the étale equivalence relation R_{Ξ} of translation. A *proper nested sequence* of tilings $T_0 \prec T_1 \prec \dots$ (each tile of T_{n+1} a patch of T_n , with buffer zones) gives a Bratteli diagram whose vertices at level n are the collared prototiles of T_n and whose edges record which tile sits in which supertile. Infinite paths are points of Ξ (the Robinson map), and tail equivalence of paths is an AF subrelation $R_{AF} \subsetneq R_{\Xi}$. The two agree on a dense G_{δ} of full measure (points that are eventually deep inside supertiles) and differ on a dense, null F_{σ} : tilings whose origin stays on supertile boundaries at every scale. To recover R_{Ξ} the authors add *horizontal* edges (adjacency of supertiles through collars) labelled by translation vectors. In dimension one this reduces to $R_{\Xi} = R_{AF} \vee \{(x_{\min}^i, x_{\max}^i)\}$, the Vershik pairs of maximal and minimal paths (Fibonacci: two pairs; Thue–Morse: four). *Essence:* the coarse-graining tower is a Bratteli diagram, generic points see only the AF part, and the full dynamics is the AF part plus a thin set of boundary identifications, which carry all the noncommutativity that the AF part misses.

Julien–Savinien [4]. For substitution tilings the boundary identifications are made combinatorial. The substitution induces substitutions ω_j on decorated j -faces for every $j < d$, and each has its own Bratteli diagram B^j . A multi-diagram joins them by horizontal edges (a j -face lies on the boundary of a $(j+1)$ -face) and by *escaping* edges (a face vanishes into the interior of a supertile). A path in B^d has *borders*: tails in B^j obtained by following horizontal edges. Its border dimension is the least such j , geometrically the dimension of $\bigcap_n \lambda^n \omega^{-n}(T)^j$. Two paths are equivalent iff they have the same border dimension j and tail-equivalent j -borders, giving $R_\Xi = \bigsqcup_j R_B^j$ with $R_B^d = R_{AF}$. Each R_B^j with $j < d$ lives on a dense null set and is not AF. *Essence*: the full relation decomposes by the dimension of the face on which a point is trapped across all scales, and each stratum is a tail relation of the face substitution of that dimension.

Putnam–Treviño [5]. A bi-infinite Bratteli diagram has levels indexed by $\mathbb{Z}$, two partial orders $\leq_s$ (edges with a common source) and $\leq_r$ (common range), and a state (ν_r, ν_s) . Here $\nu_r(v) = \sum_{r(e)=v} \nu_r(s(e))$ propagates forward, $\nu_s(v) = \sum_{s(e)=v} \nu_s(r(e))$ propagates backward, and $\sum_{v \in V_n} \nu_r(v) \nu_s(v)$ is independent of n (their Lemma 2.7). The cylinder of paths through a finite path p has measure $\nu_r(s(p)) \nu_s(r(p))$ and maps to a rectangle of width $\nu_r(s(p))$ and height $\nu_s(r(p))$. After removing extremal paths and the *singular* set $\Sigma_B = \{\Delta_s \Delta_r \neq \Delta_r \Delta_s\}$, where the two successor maps fail to commute, the quotient of the path space by both successor relations is a translation surface. Right-tail equivalence becomes the horizontal foliation and left-tail equivalence the vertical one. Their C^* -algebras are AF and foliation algebras linked by explicit subhomogeneous inductive limits. The successor flip F_B gives an even Fredholm module whose commutant inside the AF algebra is the algebra of the quotient (Theorem 9.7), and an excision argument computes the K -theory. Infinite genus appears exactly when $\Sigma_B \neq \emptyset$ (Chamanara’s surface). In finite genus the diagram comes from backward Rauzy–Veech induction, $K_0(C^*(F^+)) \cong H_1(S, \Sigma; \mathbb{Z})$, the order on K_0 is given by the Schwartzman asymptotic cycle (the top Zorich direction), and the Teichmüller flow is a deformation of the state $(e^{-t}\nu_r, e^t\nu_s)$ followed by a shift of the diagram. *Essence*: two transverse tail structures and a pairing-conserving state make a flat surface. The obstruction to flatness is the non-commutation of the two successor maps, and renormalization trades state deformation against shifts.

Giannakis–Montgomery [6]. Koopman evolution $f \mapsto f \circ \Phi^t$ is studied without a reference measure, on a reproducing-kernel Hilbert algebra H . The generator $V = S + A$ (symmetric plus antisymmetric) is dilated to a skew-adjoint tridiagonal L on $\ell^2(\mathbb{N}) \otimes H$ with A on the diagonal and $\pm S$ off it. The Fibonacci vector $Ef = \sum_i F_i e_i \otimes f$ intertwines, $LE = EV$ (because $F_{i+1} - F_{i-1} = F_i$), so the corner $(p_1 \otimes 1)L^k E_n$ reproduces V^k . A second quantization on a weighted symmetric Fock space (a Banach algebra under $\vee$ for subconvolutive weights) turns L into a free derivation D . Creation and annihilation operators are bounded, $(\mathcal{A}_n, \mathcal{F}, -iD)$ is a weak spectral triple, and $\Psi([D, a]) = V(\Psi a)$. Points of X embed as high-grade limits of truncated coherent states. The creation/annihilation commutators vanish on high grades, so the quotient by the compact diagonal ideal is commutative (the classical limit), and expectations under any Borel measure are kernel mean embeddings $\Psi_\mu = \int \Psi_x d\mu$. *Essence*: a nonlinear or dissipative dynamics becomes linear and unitary on an amplified Fock space, the classical system is the high-occupation quotient, and averages over laws are mean embeddings.

Paul [7]. For an Anosov Hamiltonian flow, the quantum evolution of an observable up to times $\hbar^{-2+\epsilon}$ (far beyond the Ehrenfest time $\log(1/\hbar)/\lambda$) is still described by a symbol. That symbol

is an element of the crossed product of $C(h^{-1}(I))$ by the group generating the strong unstable foliation, and it is obtained by pushing the initial symbol forward along the flow. Operators are written as superpositions of rank-one maps between Lagrangian states built from Gaussian coherent states along unstable leaves. Products of symbols are convolutions along leaves, and off-diagonal Husimi functions $R_\psi(z, z')$ with z' on the unstable leaf of z are the matching states. *Essence*: past the time at which a coherent state is sheared across many scales, the classical limit is the noncommutative algebra of the foliation that the shearing produces, and two-point objects take over from densities.

Francis [8]. A transverse order k foliation has finitely many leaves: one hypersurface L and the components of its complement. It is generated near L by vector fields tangent to L to order k , locally $\{y^k \partial_y, \partial_{x_i}\}$. Loops in L have holonomy only at the level of $(k-1)$ -jets, a homomorphism $\pi_1(L) \rightarrow J^{k-1}$, and this is a complete local invariant that takes all values. The holonomy groupoid is a blowup of $(M \setminus L)^2 \cup L^2$ in which L^2 is replaced by the gauge groupoid of a principal $J_{\mathbb{R}}^k$ -bundle. Its C^* -algebra is an extension $0 \rightarrow \mathcal{K}(\oplus \mathcal{K}) \rightarrow C^*(G) \rightarrow C^*(\Gamma_{\mathbb{R}}) \otimes \mathcal{K} \rightarrow 0$, and $J^2 \cong$ the $ax + b$ group. If L has a global defining function with a given $(k-1)$ -jet (Scott's b^k -manifolds), the holonomy is trivial. *Essence*: singular foliations with the same leaves are distinguished by order of tangency and jet holonomy, and the isotropy groups are solvable jet groups.

Fischer–Laurent–Gengoux [9]. Near a leaf L with transverse model $\mathcal{T}_0$ (a formal singular foliation of $\mathbb{R}^d$ at 0), singular foliations are classified formally by an outer holonomy $\pi_1(L) \rightarrow \text{Out}(\mathcal{T}_0)$ together with a finite-dimensional principal bundle, built from a Galois cover and a group extension of $\text{Inner}(\mathcal{T}_0)$. If L is contractible the foliation is the product with $\mathcal{T}_0$; if L is simply connected only the principal bundle remains; if $\mathcal{T}_0$ vanishes quadratically only the outer holonomy remains. Parallel transport along a leaf maps transverse leaves to transverse leaves (Yang–Mills connections). *Essence*: “travelling along a leaf one sees the same landscape through the window”. What can change is only the global twisting, measured by symmetries of the transverse model.

Louis [10]. For a singular foliation $\mathcal{F}$ with an anchored bundle $\rho : A \rightarrow TM$, the Nash blowup replaces each point by all limits of the kernels $\ker \rho_x$ from regular points. The pulled-back foliation becomes Debord, i.e. the image of a Lie algebroid $D_{\mathcal{F}}$ (Mohsen's groupoid). The Helffer–Nourrigat cone $\text{HN}(\mathcal{F}) = \bigcup_{x \text{ regular}} \rho_x^*(T_x^*M)$ is the image of $D_{\mathcal{F}}^*$; it is a union of symplectic leaves of the linear Poisson structure, and it is exactly where the principal symbol of a longitudinal operator is intrinsic. Longitudinal ellipticity means positivity of the symbol on $D_{\mathcal{F}}^*$. For Debord foliations the blowup is trivial and $\text{HN} = A^*$. *Essence*: the symbol space of a singular foliation is the closure of its regular conormal data, and resolving the foliation by limits of tangent spaces makes it a Lie algebroid.

3 The cocycle: a network is a weighted Bratteli diagram

Definition 3.1 (The layered diagram of an input). For x off the walls let B_x be the bi-infinite-in-spirit (here finite, levels $0, \dots, L$) weighted Bratteli diagram with $V_l = [n]$ (neurons of layer l ; V_0 the input coordinates) and edge weights $E_l(x) = D_l(x)W_l$ from V_{l-1} to V_l . A path $p = (p_0, \dots, p_L)$ has weight $\omega(p) = \prod_{k=1}^L E_k(x)_{p_k p_{k-1}}$. Put $\nu_r(l) = h_l$ and $\nu_s(l) = \delta_l$ for $f = c^\top h_L$.

Theorem 3.2 (Forward and backward states; the conserved pairing). *On the tile of x :*

1. $\nu_r(l) = E_l \nu_r(l-1)$ and $\nu_s(l-1) = E_l^\top \nu_s(l)$: activations are a left-harmonic state and sensitivities a right-harmonic state of B_x .
2. $\sum_{v \in V_l} \nu_r(v) \nu_s(v) = \langle h_l, \delta_l \rangle = f(x)$ for every l (Putnam–Treviño’s Lemma 2.7).
3. $f(x) = \sum_p x_{p_0} \omega(p) c_{p_L}$, and the paths through a fixed sub-path q from level l to level m contribute $h_{l,q_l} \omega(q) \delta_{m,q_m}$: the cylinder measure $\nu_r(s(q)) \omega(q) \nu_s(r(q))$.
4. For $s \in \mathbb{R}_{>0}^n$ the substitution $W_l \mapsto \text{diag}(s) W_l$, $W_{l+1} \mapsto W_{l+1} \text{diag}(s)^{-1}$ leaves all gates and f unchanged and acts by $h_l \mapsto s \odot h_l$, $\delta_l \mapsto s^{-1} \odot \delta_l$. It preserves every rectangle $R_{l,i} = [0, h_{l,i}] \times [0, \delta_{l,i}]$: the RELU rescaling symmetry is the diagonal (Teichmüller-type) flow $(e^{-u} \nu_r, e^u \nu_s)$, neuron by neuron.
5. Crossing the wall $Z_{l,a}$ changes E_l by $\pm e_a e_a^\top W_l$ (rank one); f is continuous and ∇f jumps by $\delta_{l,a} \nabla z_{l,a}$, the kink of stage 10.

Proof. (1) is the forward recursion and the chain rule $\delta_{l-1} = W_l^\top D_l \delta_l$. (2):

$$\langle h_l, \delta_l \rangle = \langle D_l W_l h_{l-1}, \delta_l \rangle = \langle h_{l-1}, W_l^\top D_l \delta_l \rangle = \langle h_{l-1}, \delta_{l-1} \rangle, \quad \langle h_L, \delta_L \rangle = f.$$

(3) expands the matrix product. (4): $\text{relu}(s_i z) = s_i \text{relu}(z)$ for $s_i > 0$, so z_{l+1} and all later quantities are unchanged; the chain rule gives the action on δ_l . (5): only $g_{l,a}$ changes, $z_{l,a} = 0$ on the wall so h_l is continuous, and later gates are unchanged at a generic wall point. $\square$

Each layer is therefore a presentation of the same total area $f(x)$ by n rectangles, re-cut by the cocycle E_{l+1} in passing to the next layer: widths transform by E_{l+1} and heights by $E_{l+1}^\top$. This is the algebra of zippered rectangles under Rauzy–Veech induction, where $\lambda' = \Theta^{-\top} \lambda$, $h' = \Theta h$ and $\lambda' \cdot h' = \lambda \cdot h$ [5, §12.2]. The area of rectangle $R_{l,i}$ is the gradient $\times$ input attribution of neuron (l, i) . Two differences are essential. The weights have both signs, so the cylinder “measure” is signed and the quotient of the path space is not a flat surface. For a nonnegative network ($W \geq 0$, e.g. the majorant $|W|$) Putnam–Treviño’s construction applies verbatim once orders on edges are fixed. And E_l is neither unimodular nor invertible, so the cocycle is critical rather than hyperbolic (Section 4).

Proposition 3.3 (Rauzy–Veech induction is a gated linear network). *Let π be an irreducible permutation on an alphabet $\mathcal{A}$ and let R be one step of Rauzy–Veech induction on $\lambda \in \mathbb{R}_+^{\mathcal{A}}$. The step is $\lambda \mapsto \lambda - g \lambda_{\alpha(1)} e_{\alpha(0)} - (1-g) \lambda_{\alpha(0)} e_{\alpha(1)}$ with the gate $g = \mathbf{1}\{\lambda_{\alpha(0)} > \lambda_{\alpha(1)}\}$, the sign of a linear form, and the next permutation (the wiring) depends on g . Hence $\lambda \mapsto R^k \lambda$ is piecewise linear and its linear pieces are the Rauzy cylinders, the sets of λ with a given type sequence $(g_1, \dots, g_k)$. On each cylinder the map is the inverse transpose of a product of k elementary unimodular matrices (the Rauzy–Veech cocycle). Each later gate is the sign of a linear form pulled back through the earlier steps, a bent hyperplane. So the cylinder partition is the activation tiling of a depth- k gated network with one gate per layer and gate-dependent wiring. Veech’s theorem [16] (cylinders shrink to rays for Lebesgue-almost every λ) says that this network’s gate sequence determines its input projectively.*

Proof. Direct from the definition of the step [5, §12.2]; the induction on k is the chain rule for piecewise-linear maps. $\square$

The check (Verification, item 2) runs 12 steps for $d = 4$, $\pi = (4321)$. The Jacobian on the cylinder is an integer matrix of determinant 1, perturbations staying in the cylinder obey the linear map to 2×10^{-11} , and all $2^6 = 64$ type patterns of length 6 occur. The comparison is instructive:

	Rauzy–Veech induction	He-ReLU network
gate	$\mathbf{1}\{\lambda_{\alpha(0)} > \lambda_{\alpha(1)}\}$, one per step	$\mathbf{1}\{z_{l,a} > 0\}$, n per layer
cocycle	elementary, unimodular, ≥ 0	$D_l W_l$, signed, rank $\approx n/2$
contraction of cylinders	exponential after Zorich acceleration	parabolic: chords $\sim 3\pi/(l + \frac{3}{2} \log l)$
spectrum	Forni, Avila–Viana: g positive exponents	critical: top exponent ≈ 0
order on K_0	Schwartzman cycle (top Zorich direction)	collective direction of the layer covariance
invariant measure	Veech’s Gauss measure	law of h_l under γ

4 Depth is a parabolic renormalization, and the two tiers are one flow

Let $J(\theta) = \sin \theta + (\pi - \theta) \cos \theta$ and $F(\theta) = \arccos(J(\theta)/\pi)$ on $[0, \pi]$: the angle map of the arc-cosine kernel [20], under which the angle between $h_{l-1}(x)$ and $h_{l-1}(x')$ evolves at infinite width (stage 10, Theorem *folding*).

Theorem 4.1 (Depth is a parabolic germ; Fatou coordinate). *1. F is real-analytic on $(0, \pi)$, it satisfies $F(\theta) < \theta$ on $(0, \pi]$, and*

$$F(\theta) = \theta - \frac{\theta^2}{3\pi} - \frac{\theta^3}{18\pi^2} + O(\theta^4)$$

(numerically the θ^4 coefficient is -0.00767). Thus 0 is a parabolic fixed point (multiplier 1) and $(0, \pi]$ lies in its attracting direction.

2. There is a real-analytic Φ on $(0, \pi]$, unique up to an additive constant, with $\Phi \circ F = \Phi + 1$ (the Fatou coordinate):

$$\Phi(\theta) = \lim_{N \rightarrow \infty} [\Phi_0(F^N \theta) - N], \quad \Phi_0(\theta) = \frac{3\pi}{\theta} + \frac{3}{2} \log \theta, \quad \Phi = \Phi_0 + O(\theta).$$

The coefficient $\frac{3}{2} = 1 - b/a^2$ for $F = \theta - a\theta^2 + b\theta^3 + \dots$, $a = 1/3\pi$, $b = -1/18\pi^2$, is the iterative residue (résidu itératif), the formal conjugacy invariant of the germ [18, 19].

3. For every $\theta_0 \in (0, \pi]$, $\theta_l = \Phi^{-1}(\Phi(\theta_0) + l)$ and

$$\theta_l = \frac{3\pi}{l + \frac{3}{2} \log l + \Phi(\theta_0) - \frac{3}{2} \log(3\pi) + o(1)}.$$

From orthogonal inputs ($\theta_0 = \pi/2$) the constant is $\Phi(\pi/2) - \frac{3}{2} \log 3\pi = 7.7548 - 3.3650 = 4.390$.

Proof. (1) $1 - J/\pi = (\pi(1 - \cos \theta) - (\sin \theta - \theta \cos \theta))/\pi = \theta^2 u(\theta)$ with u analytic, $u(0) = \frac{1}{2}$, and $F = 2 \arcsin \sqrt{(1 - J/\pi)/2} = 2 \arcsin(\theta \sqrt{u/2})$, which is analytic. Expanding, $\cos F = 1 - \frac{\theta^2}{2} + \frac{\theta^3}{3\pi} + \frac{\theta^4}{24} + O(\theta^5)$; matching $\cos(\theta + a_2\theta^2 + a_3\theta^3)$ gives $a_2 = -1/3\pi$ and $a_3 = -a_2^2/2 = -1/18\pi^2$.

$F < \theta$ because $J(\theta)/\pi - \cos \theta = (\sin \theta - \theta \cos \theta)/\pi > 0$ on $(0, \pi]$. (2) With $\Phi_0 = 1/(a\theta) + c \log \theta$ one computes $\Phi_0(F\theta) - \Phi_0(\theta) - 1 = ((a^2 - b)/a - ca)\theta + O(\theta^2)$. The choice $c = (a^2 - b)/a^2 = \frac{3}{2}$ makes the defect $O(\theta^2)$. Since $F^N \theta = O(1/N)$ the defects along an orbit are summable, so the limit exists, is analytic (uniform limits of analytic functions on compact subsets of the petal), satisfies $\Phi \circ F = \Phi + 1$, and differs from Φ_0 by $O(\theta)$. Uniqueness up to constants is the standard uniqueness of Fatou coordinates on a petal [18, §10]. (3) Invert $\Phi_0(\theta_l) + O(\theta_l) = \Phi(\theta_0) + l$ with $\log \theta_l = \log 3\pi - \log l + o(1)$. $\square$

Numerically (Verification, item 3) $\Phi_0(\theta_l) - l$ equals 7.388, 7.693, 7.748, 7.7542, 7.7548 at $l = 10, 10^2, 10^3, 10^4, 10^5$, while $3\pi/\theta_l - l$ keeps growing like $\frac{3}{2} \log l$. At the competition depth the two laws give

$$\theta_{16} = 0.3779 \text{ (exact)}, \quad \frac{3\pi}{16 + \frac{3}{2} \log 16 + 4.39} = 0.3839, \quad \frac{3\pi}{16} = 0.5890.$$

So the stage-10 folding law is a leading-order statement whose correction is large at $L = 16$.

Corollary 4.2 (Gate counts, corrected). *For two independent standard inputs ($\theta_0 = \pi/2$ at large n) the expected number of layer- l gates on which they differ is $n\theta_{l-1}/\pi$. The total over L layers is $\frac{n}{\pi} \sum_{k < L} \Phi^{-1}(\Phi(\pi/2) + k) = 3n \log L + O(n)$, but the $O(n)$ term is large and negative: for $L = 16$ the total is $3.66n$, not $3n \log 16 = 8.32n$ (the stage-10 prediction, which keeps only the leading term).*

Farey, not Gauss. The additive Rauzy–Veech induction for two intervals is the Farey map, which has a parabolic fixed point with infinite invariant measure. Zorich’s acceleration (grouping consecutive steps of one type) makes it the Gauss map, with a finite measure and an integrable cocycle. The He-critical network is on the Farey side: depth acts on chords by a parabolic germ, and the acceleration that turns it into a translation is precisely the Fatou coordinate. The network is not hyperbolic at any finite depth. This is the renormalization-theoretic meaning of “critical initialisation”.

Theorem 4.3 (The two tiers are one flow). *Along Eldan’s localization ($\Sigma_t = I/(1+t)$) two copies conditionally independent given $\mathcal{F}_t$ have input correlation $t/(1+t)$; at infinite width the fraction of layer- l gates on which they disagree is $\theta_{l-1}(t)/\pi$ with $\theta_0(t) = \arccos \frac{t}{1+t}$, and $\mathbb{E} \text{Var}(g_{l,a} | \mathcal{F}_t) = \theta_{l-1}(t)/2\pi$ (stage 10, Theorem depthtrickle). Let*

$$\tau(l, t) = \Phi(\theta_0(t)) + l - 1 \quad (\text{the Fatou time}).$$

1. $\mathbb{E} \text{Var}(g_{l,a} | \mathcal{F}_t) = \Phi^{-1}(\tau)/2\pi$: the unresolved gate variance depends on (l, t) only through τ . More generally every infinite-width quantity built from finitely many exchangeable conditionally independent replicas at layer l and time t depends on (l, t) only through τ .
2. The map $(t, l) \mapsto (t', l - 1)$ with $\Phi(\theta_0(t')) = \Phi(\theta_0(t)) + 1$ preserves all of them: running the upper tier from t to t' is the same as deleting one layer of the lower tier.
3. As $t \rightarrow \infty$, $\tau = 3\pi \sqrt{(1+t)/2} - \frac{3}{4} \log \frac{1+t}{2} + l - 1 + O(1)$, so the localization time worth one layer satisfies $\Delta \sqrt{1+t} \rightarrow \sqrt{2}/3\pi = 0.1500$.

Proof. (1) $\theta_{l-1}(t) = F^{l-1}(\theta_0(t)) = \Phi^{-1}(\Phi(\theta_0(t)) + l - 1)$ by Theorem 4.1. At infinite width the preactivations of the replicas at layer l are jointly Gaussian with common variance (independent of t , since $X \sim \gamma$ marginally) and common pairwise correlation $\cos \theta_{l-1}(t)$ (the dual-kernel recursion applied to each pair). (2) follows from (1). (3) $1 - \cos \theta_0 = 1/(1+t)$ gives $\theta_0 = \sqrt{2/(1+t)} (1 + O(1/t))$; insert into $\Phi = \Phi_0 + O(\theta)$. $\square$

Putnam–Treviño’s Teichmüller flow, realized. In [5, §12.8] the Teichmüller flow on bi-infinite diagrams built from zippered rectangles is the identification

$$(B, e^{-t_R} \nu_r, e^{t_R} \nu_s) \sim (\sigma B, \sigma_* \nu_r, \sigma_* \nu_s).$$

It deforms the state for a renormalization time t_R and then shifts the diagram by one level. Theorem 4.3(2) is this identification for the two-tier network. The diagram is the depth tower, its state is the gate law conditional on $\mathcal{F}_t$ (the upper tier), the shift removes a layer, and the renormalization time is read off the Fatou coordinate. In log-time a layer costs $\Delta \log(1+t) \approx 2\sqrt{2}/(3\pi\sqrt{1+t})$: it becomes cheaper as the localization proceeds.

Data collapse. The stage-10 measured disagreement fractions (width 512, 64 pairs, three times, six layers), sorted by τ (Verification, item 4):

τ	t	l	meas.	pred.	τ	t	l	meas.	pred.
9.80	1	1	.333	.333	24.80	1	16	.136	.115
10.80	1	2	.287	.292	27.02	9	8	.106	.105
12.80	1	4	.223	.236	31.02	9	12	.082	.092
16.80	1	8	.174	.173	35.02	9	16	.098	.081
20.02	9	1	.140	.144	63.76	99	1	.044	.045
20.80	1	12	.130	.138	70.76	99	8	.040	.041
21.02	9	2	.135	.136	74.76	99	12	.033	.039
23.02	9	4	.116	.124	78.76	99	16	.043	.037

Points from different localization times interleave on a single decreasing curve, e.g. $(t, l) = (9, 1)$ at $\tau = 20.0$ and $(1, 12)$ at $\tau = 20.8$. The three last-layer points (bold) all sit above it. This is the finite-width drift of the last layer that stage 10 recorded, and the collapse makes it a clean diagnostic: a mean-field (replica-symmetric, infinite-width) quantity cannot leave the τ -curve, so the excess at $l = 16$ is a finite-width effect.

Proposition 4.4 (Tail equivalence at infinite width). *For $\theta_0 > 0$, $\sum_l \theta_l = \infty$ (harmonic, from Theorem 4.1). At infinite width two distinct inputs therefore disagree on a positive fraction θ_{l-1}/π of the gates of every layer, and the total expected disagreement diverges even per neuron. The gate sequences of distinct inputs are never tail-equivalent.*

Conjecture 4.5 (Finite width is a hyperbolic perturbation). *At width n the projective contraction of products of gated Gaussian matrices adds a linear term, $\theta \mapsto (1 - c_n)\theta - \theta^2/3\pi + \dots$ with $c_n \asymp 1/n$ (the gap between the top two Lyapunov exponents). The parabolic point becomes attracting-hyperbolic, the crossover depth is $\asymp n$, $\sum_l \theta_l < \infty$, and almost every pair of inputs (on non-dead networks) becomes eventually tail-equivalent. This would settle the stage-10 question about the tail algebra of the depth tower: it is decided by the summability of the Fatou orbit. The competition networks ($L = 16 \ll n = 1024$) sit deep in the parabolic regime.*

5 The walls as a stratified singular foliation

Work on $\mathbb{R}^n \setminus \{0\}$; everything is conical. The depth- L tiling stratifies input space into relatively open polyhedral cones: open tiles, open pieces of walls, and so on. These are the *faces*. Each face is convex, hence contractible.

Definition 5.1 (Wall foliation). The wall foliation $\mathcal{F}_L$ is the module of (formal) vector fields tangent to every wall $Z_{l,a}$, $l \leq L$. Its leaves are the faces. At a point of a face S of codimension j the *transverse model* is the induced formal foliation of a j -dimensional slice.

The radial (Euler) field $x \cdot \nabla$ is tangent to every wall, because walls are cones. So the drift of the Ornstein–Uhlenbeck generator $\Delta - x \cdot \nabla$ is longitudinal and only the diffusion is transverse. The Euler–Stein identity $\mathbb{E}f = \mathbb{E}[x \cdot \nabla f] = \mathbb{E}[\Delta f]$ for 1-homogeneous f records where the longitudinal part meets the transverse one.

Theorem 5.2 (Normal and bent crossings). *Let S be a codimension-two face at a generic point of which exactly two walls meet, $Z = Z_{l,a}$ and $Z' = Z_{k,b}$ with $k \leq l$. Let $T = \partial z_{l,a} / \partial h_{k,b}$ be the gated transport coefficient (intermediate gates fixed at the point; $T = 0$ if $k = l$).*

1. *If $T = 0$ the walls are transverse hyperplanes near S . The transverse model is the log foliation of $y_1 y_2 = 0$, generated by $y_1 \partial_1$ and $y_2 \partial_2$; the isotropy Lie algebra is $\mathbb{R}^2$ (abelian). The foliation is Debord near S (its algebroid is the log tangent bundle), its Nash blowup is trivial, and its Helffer–Nourrigat cone is the whole log cotangent bundle [10]. The holonomy groupoid is the b -groupoid of Melrose’s calculus [14].*
2. *If $T \neq 0$, choose transverse coordinates $y_1 = z_{k,b}$ and $y_2 =$ the linear function equal to $z_{l,a}$ on the side $y_1 < 0$. Then $z_{l,a} = y_2 + T \operatorname{relu}(y_1)$ near S , so $Z = \{y_2 = 0, y_1 \leq 0\} \cup \{y_2 + T y_1 = 0, y_1 \geq 0\}$ is bent along Z' . The formal transverse model is the log foliation of the central three-line arrangement*

$$y_1 y_2 (y_2 + T y_1) = 0,$$

a free module (Saito) with basis $E = y_1 \partial_1 + y_2 \partial_2$ and $\vartheta_T = -y_1(y_2 + T y_1) \partial_1$. The isotropy Lie algebra is $\operatorname{span}(E, \vartheta_T)$ with $[E, \vartheta_T] = \vartheta_T$, i.e. $\mathfrak{aff}(1)$, non-abelian, and the linear isotropy is $\mathbb{R}E$, of dimension 1 instead of 2.

3. *The integrated isotropy at a bent crossing is the connected $ax + b$ group $\operatorname{Aff}_+(1)$, which Francis identifies with the jet group J^2 [8]. The groupoid of $\mathcal{F}_L$ restricted to S is $S \times S \times \operatorname{Aff}_+(1)$, with C^* -algebra $C^*(\operatorname{Aff}_+(1)) \otimes \mathcal{K}$, an extension of $C_0(\mathbb{R})$ by $\mathcal{K} \oplus \mathcal{K}$. At a normal crossing it is $C_0(\mathbb{R}^2) \otimes \mathcal{K}$.*

Proof. Near a generic point of S only the gate $g_{k,b}$ changes, and crossing Z' toggles $h_{k,b}$ between 0 and $z_{k,b}$, which changes $z_{l,a}$ by $T z_{k,b}$. This gives the normal form. A formal vector field is tangent to a half-plane through S iff each homogeneous component is, and a homogeneous polynomial field tangent to a half-line in the slice is tangent to the whole line (polynomial identity on a ray). So the formal module is the module of fields tangent to the three lines $\{y_1 = 0\}$, $\{y_2 = 0\}$, $\{y_2 + T y_1 = 0\}$; the extensions of the two half-lines of Z are *phantom* lines that the formal structure cannot distinguish from the wall. Saito’s criterion holds: $\det(E, \vartheta_T) = y_1 y_2 (y_2 + T y_1)$. $[E, \vartheta_T] = (2 - 1)\vartheta_T$ because ϑ_T is homogeneous of degree 2, and $\vartheta_T \notin I_0 \mathcal{F}$ (otherwise $\vartheta_T = \ell E$ with ℓ linear and the determinant would vanish). A linear field tangent to three distinct lines through 0 has three eigenlines, hence is scalar. For $T = 0$ two of the lines coincide and the reduced arrangement is $y_1 y_2 = 0$. (3): the free module is the section module of a Lie algebroid with injective anchor on the regular part, so the holonomy groupoid is the Lie groupoid integrating it [11, 10]; the simply connected group with Lie algebra $\mathfrak{aff}(1)$ is $\operatorname{Aff}_+(1)$, which has trivial centre. The structure of $C^*(\operatorname{Aff}_+(1))$ is classical. $\square$

The check (Verification, item 5) solves the tangency conditions for homogeneous polynomial fields in the slice. The dimensions in degrees 1, 2, 3 are 2, 4, 6 at $T = 0$ and 1, 3, 5 at $T = 0.3$, and $[E, \vartheta_T] = \vartheta_T$ holds symbolically.

The bent crossings are the memory. T is the coefficient with which the kink of neuron (k, b) enters the deep pre-activation: the coupling at which Price’s theorem gives birth to the third cumulant (stage 10, Theorem *birth*). In the simplest case it is literally the cumulant. For $z_{2,a} = \sum_b W_2(a, b) \text{relu}(w_b \cdot x)$ with orthogonal first-layer rows, the $w_b \cdot x$ are independent, the bending coefficients are $T_{ab} = W_2(a, b)$, and

$$\kappa_3(z_{2,a}) = c_3 \sum_b T_{ab}^3 |w_b|^3, \quad c_3 = \kappa_3(\text{relu } Z) = \frac{\pi + 2}{(2\pi)^{3/2}} = 0.3265.$$

So $z_{2,a}$ is Gaussian iff its wall is nowhere bent, and its skewness is the sum of cubed bendings over its bent crossings. With correlated rows, cross terms appear, weighted by the codegrees (cosines) of the normal crossings among the shallow walls. These are the walk sums with exact contacts of stage 10, now read as an expansion over the codimension-two strata of the wall foliation: bent crossings are vertices with a contact, normal crossings are edges. The Gaussian closure is exact for any wall system with only normal crossings, and every family of jointly Gaussian pre-activations is the pre-activation family of such a system in an augmented Gaussian space (realize the covariance as $\tilde{z}_l = M_l \xi$). Its error is therefore carried by the bent crossings.

Smooth versus piecewise linear. The piecewise-linear homeomorphism $(y_1, y_2) \mapsto (y_1, y_2 + T \text{relu}(y_1))$ unbends Z into a normal crossing. In the PL category every crossing is normal. The bending, the phantom line and the non-abelian isotropy exist only for the smooth (formal) structure, and the Gaussian measure is a smooth object, not a PL one. This is the geometric content of “Gaussian calculus sees the kinks”: non-Gaussianity is the obstruction to smoothing the unbending chart.

Proposition 5.3 (No holonomy). *(1) Every face is contractible, so by Fischer–Laurent–Gengoux ([9], Cor. 3.10) the formal wall foliation along any face is the product of the face with its transverse model: outer holonomy and principal-bundle data are trivial. (2) Every wall $Z_{l,a}$ has the global defining function $z_{l,a}$; for each k the transverse-order- k structure it induces is Scott’s b^k structure with a global $(k-1)$ -jet [15], whose holonomy invariant is trivial ([8], §1). Hence the noncommutativity of the wall foliation is purely isotropic: abelian at normal crossings, $\mathfrak{aff}(1)$ at bent crossings, and solvable at higher-codimension faces.*

The contrast with tilings is instructive. In a substitution tiling the boundary relations R_B^j of Julien–Savinien are large: dense and of tail type. In the network every face is a convex cone and the global defining functions kill all holonomy. Whatever the network has that is genuinely noncommutative lives at points (isotropy) and in the depth-and-localization dynamics (Sections 4, 6), not in the leaf space.

6 The upper tier as a semiclassical deformation

The upper tier is the family of Gaussian laws $N(m, \hbar I)$, $\hbar \in (0, 1]$; Eldan’s path has $\hbar = 1/(1+t)$ and a moving centre. For a network output f put

$$F(m, \hbar) = \mathbb{E} f(m + \sqrt{\hbar} Z), \quad \partial_{\hbar} F = \frac{1}{2} \Delta_m F, \quad u = F(0, 1).$$

In Paul’s language $N(m, \hbar I)$ is the squared modulus of a Gaussian coherent state of width $\sqrt{\hbar}$, and $F(\cdot, \hbar)$ is the Husimi (anti-Wick, Toeplitz lower) symbol of the operator of multiplication by f . The classical limit $\hbar \rightarrow 0$ returns f , which is singular along the wall foliation.

Theorem 6.1 (Boundary layer; the parabolic blowup). *Let m lie within distance ρ of a single flat wall piece W of f , with all other walls at distance at least $R > \rho$, signed distance d to W and kink κ (the jump of the normal derivative of f across W). Then*

$$F(m, \hbar) = f(m) + \sqrt{\hbar} \kappa \psi\left(\frac{d}{\sqrt{\hbar}}\right) + O(e^{-(R-\rho)^2/2\hbar}),$$

$$\psi(u) = \varphi(u) - |u| \Phi_{\mathcal{N}}(-|u|) = \mathbb{E}(u + Z)^+ - u^+.$$

Hence F lifts to a smooth function on the parabolic blowup of $W \times \{0\}$ in $(m, \hbar)$ -space, with radial coordinate $r = (d^2 + \hbar)^{1/2}$ and angular coordinate $d/\sqrt{\hbar}$. The correction is homogeneous of degree one in r , i.e. of transverse order one in Francis's sense. Near a codimension- j face, iterated parabolic blowups are needed, performed deepest stratum first: the combinatorics of the resolution is the face multi-diagram of Section 7.

Proof. On the ball of radius R about the foot point, $f = \ell + \kappa d^+$ with ℓ linear. $\mathbb{E}\ell(m + \sqrt{\hbar}Z) = \ell(m)$ and $\mathbb{E}(d + \sqrt{\hbar}Z_\nu)^+ = \sqrt{\hbar}(\varphi(u) + u\Phi_{\mathcal{N}}(u))$ with $u = d/\sqrt{\hbar}$; subtract d^+ . The contribution of the complement of the ball is bounded by the Gaussian tail times a linear bound on $|f|$. $\square$

The check (Verification, item 6) uses a two-dimensional network with six first-layer walls and a base point 3.9 from the other walls. For $\hbar \in \{0.01, 0.05, 0.2, 0.5\}$ and $d/\sqrt{\hbar} \in \{0, 0.7\}$ the formula holds to relative error 10^{-8} , the quadrature accuracy. Stage 10's Euler–Stein constant reads naturally on this blowup. A defect of Hermite contact order d is homogeneous of degree $d + 1$ in r (transverse order $d + 1$), and averaging over a uniformly distributed radius gives $\int_0^1 r^{d+1} dr = 1/(d + 2)$.

Proposition 6.2 (The target is deep in the quantum regime; the right limit is kinetic). (1) *At scale $\hbar$ the expected number of unresolved first-layer gates is $\mathbb{E}\sum_a \text{Var}(g_{1,a} \mid \text{ball}) = n\theta_0(\hbar)/2\pi$ with $\theta_0 = \arccos(1 - \hbar)$. At the target $\hbar = 1$ this is $n/4$, and the single-wall (semiclassical) regime $n\theta_0/2\pi \lesssim 1$ requires $\hbar \lesssim 2\pi^2/n^2 \approx 1.9 \times 10^{-5}$ for $n = 1024$. (2) Each wall bends the downstream pre-activations by $T = O(n^{-1/2})$ and there are n walls per layer with $\sum_b T_{ab}^2 g_b \approx 1$ at He initialisation: coupling² $\times$ density $= O(1)$. This is the weak-coupling (van Hove) scaling of a particle among many weak scatterers [21]. The infinite-width (NNGP, moment-chain) description is the kinetic limit of this scaling, not a semiclassical expansion, and the finite-width memory is its correction. Ehrenfest-type arguments therefore cannot control the memory.*

Proof. (1) is stage 10's unresolved-variance formula at layer one with correlation $1 - \hbar$; at $\hbar = 1$ each gate is a fair coin. For small $\hbar$, $\theta_0 \approx \sqrt{2\hbar}$. (2) $W_2(a, b) \sim N(0, 2/n)$ and half the gates are open. $\square$

Paul's two-point symbols. Past the Ehrenfest time Paul's symbols become kernels $\sigma(z, z')$ along unstable leaves, and the matching states are off-diagonal Husimi functions $R_\psi(z, z')$. The network's counterparts are stage 10's time-correlated Kikuchi weights $K_{l,t}(a, b) = \mathbb{P}(g_{l,a}(X) = 1, g_{l,b}(X') = 1)$ and the angle operator $\Theta_{l,t} = E_{\mathfrak{B}_l} E_{\mathfrak{F}_t} E_{\mathfrak{B}_l}$. Both are built from two coherent samples X, X' of one localized law, so they are two-point objects of exactly Paul's kind. In the network the role of the unstable foliation (the direction in which the coherent state is sheared) is played by the folding of the depth tower, whose clock is the Fatou time. Theorem 4.3 says these two-point objects are functions of τ alone at infinite width. (Analogy, with the exact content of Theorem 4.3.)

Giannakis–Montgomery’s Fock space.

1. The Wiener chaos of γ is the symmetric Fock space over $\mathbb{R}^n$. Exponential vectors are the tilted densities $e^{m \cdot x - |m|^2/2}$ (coherent states), with $a_v e^{m \cdot x - |m|^2/2} = \langle v, m \rangle e^{m \cdot x - |m|^2/2}$: localization centres are coherent states. A moment chain of order K is a truncation to $\leq K$ quanta. The linear part of a layer acts by second quantization (k -th moment tensors transform by $W^{\otimes k}$, the functor $\Gamma(W)$), and the RELU is a vertex that mixes grades.
2. Depth drives the network toward the commutative quotient. Stage 10’s Mehler pair temperature gives a mean token number $\approx 1/(1 - \cos \theta_{l-1})$ for a neuron pair at layer l , so the effective Planck constant is $\hbar_l \approx 1 - \cos \theta_{l-1} \approx \theta_{l-1}^2/2 \approx 9\pi^2/2(l + \frac{3}{2} \log l + 4.39)^2$ by Theorem 4.1. On high Fock grades the weighted canonical commutation relations become commutative ([6], Prop. 32), which is the classical limit.
3. The target is a kernel mean embedding: $u = \mathbb{E}_\gamma h_L$ is the mean embedding of γ for the random feature map h_L , with kernel $k_L(x, x') = h_L(x) \cdot h_L(x')/n$ (exact). For any surrogate law ν , $\frac{1}{n} \|\mathbb{E}_\gamma h_L - \mathbb{E}_\nu h_L\|^2 = \text{MMD}_{k_L}^2(\gamma, \nu)$ [22]. The competition’s mean squared error of such an estimator is a maximum mean discrepancy in the depth- L feature space.
4. The Fibonacci dilation realizes a non-normal generator as the corner $(p_1 \otimes 1)L^k E_n$ of a skew-adjoint banded operator on $\ell^2(\mathbb{N}) \otimes H$. Stage 10 realized the network as the corner $[(1 - D_\downarrow P_x)^{-1}]_{L0}$ of a resolvent of a banded chiral operator on $\ell^2(\{0, \dots, L\}) \otimes \mathbb{R}^n$. A unitary dilation of the gated transfer, which would make the layer law’s Gaussian part invariant (a Koopman–von Neumann wavefunction for the network), is open.

7 Borders: the mean lives on the codimension-one faces

Proposition 7.1 (Two Bratteli diagrams of the depth tower). *1. Tile diagram. The vertices at level l are the nonempty depth- l tiles, edges are inclusions, and $\nu(v) = \gamma(v)$ is a one-sided state ($\nu(v) = \sum_{\text{children}} \nu$). It is a tree, so its AF algebra is commutative. On a great circle, with the angular order as $\leq_s$, it is Putnam–Treviño’s “decimal expansion” picture: the circle is the quotient of the path space by successor identifications, and the successor pairs are the two sides of the breakpoints.*

2. Fiber diagram. The supertiles of level l are the fibres of h_l ; write $x \approx_l x'$ iff $h_l(x) = h_l(x')$. These relations increase with l (the AF direction), their classes are polyhedra of dimension $n - \text{rank}(D_l W_l \cdots D_1 W_1)$ (about $n/2$ from the first layer on), and $\approx_\infty = \bigcup_l \approx_l$ (eventual indistinguishability) is contained in the tail relation of the letter diagram, whose level- l vertices are the layer- l gate patterns alone (finite rank 2^n). The letter diagram’s tail relation is the network’s R_{AF} .

3. Borders. Call $n - j$ the border dimension of x if, for every large depth, x lies on a face of codimension j . At infinite width and infinite depth every Gaussian ball, of any radius $\sqrt{\hbar} > 0$, meets walls of every layer, with $\sum_l n \theta_{l-1}(\hbar)/2\pi = \infty$ by Theorem 4.1. So points of border dimension $< n$ are dense; they are null. This is Julien–Savinien’s thin set (their Prop. 3.22 and Rem. 3.26), and it is dense precisely because depth is parabolic: under the hyperbolic finite-width dynamics of Conjecture 4.5 the sum converges.

4. Births. A wall of layer l inside a tile of depth $l - 1$ is a face that does not descend from a face of the parent. The face multi-diagram (Bratteli diagrams of faces of each codimension,

joined by incidence edges) therefore has birth edges from tiles at level $l-1$ to codimension-one faces at level l . These are the reverse of Julien–Savinien’s escaping edges (faces swallowed by supertiles), and they are where stage 10 locates the creation of contacts.

Proof. (1) and (2) are immediate from the definitions; $h_l(x) = h_l(x')$ forces equal gates at all later layers. (3) uses the infinite-width unresolved variance at every layer and the divergence of $\sum \theta_l$. (4) is the definition of a wall. $\square$

Proposition 7.2 (The mean is a border functional). *For any network output f (piecewise linear, 1-homogeneous),*

$$\mathbb{E}_\gamma f = \mathbb{E} \Delta f = \frac{1}{\sqrt{2\pi}} \sum_F \kappa_F \alpha_F,$$

where F runs over the codimension-one faces of the depth- L tiling, κ_F is the jump of the normal derivative across F , and α_F is the Gaussian measure of the cone F inside its hyperplane (its solid-angle fraction). Equivalently, let $H_\partial = L^2(\bigcup_F F \times \{+, -\}, \varphi dS)$ (two copies of each wall piece, one per side), let Γ be the side grading, $\mathbb{F}$ the side flip, and $\pi(g)$ multiplication by the one-sided values of a field g that is constant on open tiles. Then

$$[\mathbb{F}, \pi(g)] = 0 \iff g \text{ is continuous across every wall,} \quad \mathbb{E}_\gamma f = \frac{1}{2} \text{Tr}_\varphi (\Gamma \mathbb{F} [\mathbb{F}, \pi(\partial_\nu f)]).$$

Proof. Euler and Stein give $\mathbb{E} f = \mathbb{E}[x \cdot \nabla f] = \mathbb{E} \Delta f$, and $\Delta f = \sum_F \kappa_F \delta_F$ as a distribution. Moreover $\int_F \varphi_n dS = \varphi_1(0) \alpha_F$ because F is a cone through 0. On one face, with $\pi(g) = \text{diag}(g_+, g_-)$, $\Gamma \mathbb{F} [\mathbb{F}, \pi(g)] = \text{diag}(g_+ - g_-, g_+ - g_-)$; take $g = \partial_\nu f$. $\square$

The second form is Putnam–Treviño’s Fredholm module with the successor flip replaced by the wall flip. Their Theorem 9.7 (functions descend to the quotient iff they commute with the flip) becomes continuity across walls, and the Gaussian mean of the network becomes the supertrace pairing of the flip with the normal derivative: the kink current of stage 10. On a continuum of wall points the commutators are multiplication operators rather than compact ones, so this is a Gaussian-weighted analogue, not a Fredholm module in the strict sense. Its restriction to a great circle, which has finitely many breakpoints, is one. The two-dimensional check (Verification, item 7) has 70 breakpoints and gives $\mathbb{E} f = 0.36919251$ both by angular quadrature and by the border sum. In Julien–Savinien’s terms, the whole mean is carried by the border paths of the codimension-one face diagram: the generic (AF) points contribute nothing.

8 What this explains, predicts, and leaves open

8.1 What the placement explains

1. **Why the folding law needed a correction.** Depth is a parabolic germ with iterative residue $3/2$, so $3\pi/l$ misses a $\frac{3}{2} \log l$ and a constant that are large at $l = 16$ (Theorem 4.1). The He-critical network is Farey-like, not Gauss-like.
2. **Why localization resolves deep gates first, quantitatively.** Depth and localization time are two coordinates of one renormalization flow; one layer is worth $\sqrt{2}/3\pi$ of $\sqrt{1+t}$ (Theorem 4.3). This is the Teichmüller flow of Putnam–Treviño: a state deformation followed by a shift of the diagram.

3. **Where the memory lives geometrically.** At bent crossings of the wall foliation, where the isotropy is $\mathfrak{aff}(1)$ instead of abelian and the bending is the gated transport coefficient (Theorem 5.2). The Gaussian closure is exact for wall systems with only normal crossings.
4. **Why tiling-theoretic holonomy is absent.** Faces are convex and walls have global defining functions (Proposition 5.3). The noncommutativity that tiling theory finds in leaf spaces is, for the network, in isotropy and in the two-tier dynamics.
5. **Why semiclassical expansions cannot reach the target.** At $\hbar = 1$ there are $n/4$ unresolved gates per layer; the infinite-width description is a kinetic (van Hove) limit (Proposition 6.2).
6. **Why the mean is a boundary quantity.** It is the border functional of the codimension-one face diagram, or the supertrace pairing of the wall flip with the normal derivative (Proposition 7.2).

8.2 Predictions (falsifiable)

1. Two independent standard inputs differ on $n\theta_{l-1}/\pi$ gates of layer l , where $\theta_1 = 1.247$, $\theta_2 = 1.054$, and so on down to $\theta_{16} = 0.378$; the total over 16 layers is $3.66n$. This replaces stage 10's $3n \log L = 8.32n$ (Corollary 4.2).
2. Every infinite-width two-tier quantity collapses on the Fatou time τ . The departure of finite-width measurements from the τ -curve is a finite-width (memory) signal; it is visible at the last layer for all three localization times in the stage-10 data.
3. With orthogonal first-layer rows, $\kappa_3(z_{2,a}) = c_3 \sum_b W_2(a, b)^3 |w_b|^3$ exactly, and $z_{2,a}$ is Gaussian iff no crossing is bent.
4. (Conjectural) At width n the parabolic regime ends at depth $\asymp n$. Beyond it the angle decays exponentially, the walls met by a small ball become finite in number, and gate sequences become eventually tail-equivalent (Conjecture 4.5).

8.3 Open problems

1. **The finite-width linear term.** Compute the projective Lyapunov gap of products of gated He matrices, prove Conjecture 4.5, and decide the tail algebra of the letter diagram (the stage-10 question).
2. **The error of the Gaussian closure as a sum over bent crossings.** Interpolate each bent crossing to a normal one (send its T to 0) and write the closure error as a sum of crossing-local terms. This is the foliation form of stage 10's contact-resolved estimator problem, and the $\mathfrak{aff}(1)$ isotropy suggests that each term is a commutator.
3. **K -theory and index.** The wall-foliation algebra has a finite composition series over the faces with subquotients $C^*(\text{isotropy}) \otimes \mathcal{K}(C_0(\mathbb{R}^j))$ at normal crossings, $C^*(\text{Aff}_+(1))$ at bent ones). Is there an index pairing, sharper than the supertrace formula, whose value is the mean or one of its chain corrections? Savinien–Bellissard's spectral sequence for tilings is the template.
4. **Nonnegative networks as flat surfaces.** For $W \geq 0$ (e.g. the majorant $|W|$) Putnam–Treviño's construction yields a genuine translation surface for each input, with widths h , heights δ and area f . Do its Teichmüller flow and asymptotic cycle bound the signed network?

5. **A Koopman–von Neumann wavefunction.** A unitary dilation of the gated transfer in the spirit of Giannakis–Montgomery, under which the Gaussian part of the layer law is invariant.
6. **Chains in Fatou time.** Stage 9 and stage 10 transport Euler–Stein constants and window corrections layer by layer. If their depth dependence is a function of τ rather than of l , one calibration at one depth fixes all depths.

Verification

The checks are run by the script `verify_stage11.py`; its output is saved next to this note.

identity	result
1. $\langle h_l, \delta_l \rangle = f$ at every layer; rescaling	$n = 64, L = 8$: max error 2.9×10^{-15} ; per-neuron rescaling changes f by 4×10^{-16} and the rectangles by 9×10^{-16}
2. Rauzy–Veech as a gated linear map	$d = 4, \pi = (4321)$, 12 steps: integer Jacobian, determinant 1; 186/200 perturbations stay in the cylinder and obey the linear map to 2×10^{-11} ; all 64 type patterns of length 6 occur
3. Parabolic germ, Fatou coordinate	$(F - \text{series})/\theta^4 = -0.00766, -0.00767$ at $\theta = 10^{-2}, 10^{-3}$; $\Phi_0(\theta_l) - l = 7.388, 7.693, 7.748, 7.7542, 7.7548$ at $l = 10, \dots, 10^5$; the $3\pi/l$ law errs by 84%, 11%, 1.5% at $l = 10, 10^2, 10^3$
4. Two-tier collapse	stage-10 width-512 data monotone in τ except the three last-layer points; $\Delta\sqrt{1+t}$ per layer = 0.1533, 0.1511, 0.1504 at $t = 99, 999, 9999$ (limit $\sqrt{2}/3\pi = 0.1500$)
5. Bent crossing	tangent fields of degree 1, 2, 3: 2, 4, 6 at $T = 0, 1, 3, 5$ at $T = 0.3$; $[E, \vartheta_T] = \vartheta_T$ symbolically
6. Boundary layer	relative error $2\text{--}8 \times 10^{-8}$ for $h \in \{0.01, 0.05, 0.2, 0.5\}$, $d/\sqrt{h} \in \{0, 0.7\}$
7. Mean as a border functional	$\mathbb{R}^2$ input, 16 + 16 hidden units, 70 breakpoints: 0.36919251 by quadrature and by the border sum

The exact folding angles from $\theta_0 = \pi/2$ are $\theta_1, \dots, \theta_{16} = 1.247, 1.054, 0.921, 0.822, 0.744, 0.681, 0.628, 0.584, 0.546, 0.513, 0.483, 0.458, 0.435, 0.414, 0.395, 0.378$, and $\sum_{l=1}^{16} \theta_{l-1}/\pi = 3.658$.

References

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- [2] Stage 9 note: *the mean of a random ReLU network as a central localization* ([notes/stage9](#)).
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